

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Third Semester B.E. (Mechanical) (Credit System) Degree Examinations**

Supplementary - April 2023

20ME303 / 17ME303 - BASIC THERMODYNAMICS

Duration: 3 Hours

Max. Marks: 100

- Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
 2) Use of **thermodynamic data handbook** may be permitted.

Unit – I**Marks BT* CO* PO***

1. a) Differentiate the following.
 - i) Macroscopic and microscopic approach
 - ii) Real boundary and imaginary boundary
 - iii) Thermal equilibrium and thermodynamic equilibrium
 - iv) Control volume and control surface
 - v) Ice point and steam point

10 L*1 1 1
- b) A new scale N of temperature is divided in which the ice point is assigned 100°N and the steam point is assigned 400°N . Establish the relation between the N scale and the Celsius scale. At what temperature will both the Celsius and the new thermometer readings be identical numerically?

10 L3 1 2
2. a) State zeroth law of thermodynamics. Explain its importance in temperature measurement.

4 L1 1 1
- b) What is Quasi-static process? What is its significance?

4 L1 1 1
- c) A Celsius scale and a Fahrenheit scale are immersed in a fluid. Calculate the temperature of the fluid, when
 - i) The numerical readings are identical in both the thermometers.
 - ii) The Fahrenheit reading numerically twice that of Celsius reading.

Also find the values in Rankine and Kelvin scale.

12 L3 1 2

Unit – II

3. a) Define the following. i) Critical point ii) Triple point iii) Dryness fraction

6 L1 2 1
- b) A vessel of volume 0.04 m^3 contains a mixture of saturated water and steam at a temperature of 250°C . The mass of the liquid present is 9 kg. Find the pressure, mass, specific volume, enthalpy, and internal energy.

8 L3 2 2
- c) A storage vessel of 0.1 m^3 contains 0.03 kg of gas at 34.33 N/cm^2 and 27°C . Determine i) Gas constant ii) Molecular weight iii) Number of moles of gas iv) Molar volume

6 L3 2 2
4. a) Explain the following. i) Compressibility factor ii) Law of corresponding states iii) Generalized compressibility chart

6 L1 2 1
- b) Draw the p-T and T-S diagram for pure substance indicate the following.
 - i) Critical point ii) Triple point iii) Sublimation curve
 - iv) Solid plus vapor region v) Liquid plus vapor region

8 L1 2 1
- c) A spherical balloon 10 m in diameter is filled with Hydrogen at 20°C and atmospheric pressure. The surrounding air is at 15°C , and barometer reads 750 mm of Hg. Determine the load lifting capacity of the balloon.

6 L3 2 2

Unit – III

5. a) Define work and heat with reference to the thermodynamic point of view. State the similarities and dissimilarities between them.

10 L1 3 1

P.T.O.

b)	Derive an expression for displacement work in case of a closed system undergoing reversible adiabatic quasi-static process.	4	L2	3	1
c)	An engine cylinder of diameter 22.5 cm has a stroke length of 37.5 cm. The swept volume is 4 times the clearance volume. The pressure of gases at the beginning of expansion stroke is 1569 kPa. Find the work done during expansion stroke assuming the process as reversible adiabatic, take $\gamma = 1.4$.	6	L3	3	2
6. a)	Differentiate point and path functions with examples.	4	L1	3	1
b)	Derive an expression for heat transfer in polytropic process.	6	L2	3	1
c)	A fluid at 0.7 bar occupying 0.09m^3 is compressed reversibly to a pressure of 3.5 bar according to a law $p v^n = \text{constant}$. The fluid is then heated reversibly at constant volume until the pressure is 4 bar; the specific volume is then $0.5\text{ m}^3/\text{kg}$. A reversible expansion according to a law $p v^2 = \text{constant}$, restores the fluid to its initial state. Sketch the cycle on a p-v diagram and calculate. (i) the mass of fluid present (ii) the value of n in the first process (iii) the network of the cycle.	10	L3	3	2
Unit – IV					
7. a)	State first law of thermodynamics for a closed system undergoing a non-cyclic process and hence prove that energy is a property of the system.	6	L1	4	1
b)	Define specific heat at constant volume and constant pressure.	4	L1	4	1
c)	A system contains 0.15 m^3 of air at a pressure of 3.8 bar and 150°C . It is expanded adiabatically till the pressure falls to 1 bar. The gas is then heated at a constant pressure till its enthalpy increases by 70 kJ. Determine the total work.	10	L3	4	2
8. a)	Write the S.F.E.E. and modify for the following cases. i) Adiabatic horizontal nozzle with negligible inlet velocity ii) An adiabatic steam turbine with negligible change in kinetic and potential energies.	10	L1	4	1
b)	A centrifugal pump delivers 50kg of water per second. The inlet and outlet pressure are 1 bar and 4.2 bar respectively. The suction is 2.2m below the center of the pump and delivery is 8.5m above the center of the pump. The suction and delivery pipe diameters are 20cm and 10cm respectively. Determine the capacity of electric motor to run the pump.	10	L3	4	2
Unit – V					
9. a)	State and prove Carnot theorem.	10	L1	5	1
b)	A household refrigerator maintains a space at a temperature of 2°C . Every time the door is opened, warm material is placed inside, introducing an average 420 KJ of heat, but making only a small change in temperature of the refrigerator. The door is opened 20 times a day and the refrigerator operate at 15 % of ideal COP. The cost of work is Rs. 2 per KW-hr. What is the monthly bill for this refrigerator? The atmosphere is at 30°C .	10	L3	5	2
10. a)	With the help of first law and second law for a process derive an expression $T dS = dU + PdV$ and $\delta Q = dH - VdP$, and hence derive an expression for entropy change of an ideal gas	10	L2	5	1
b)	1.2m^3 of air is heated reversibly at constant pressure from 300K to 600K and is then cooled reversibly at constant volume back to initial temperature. If the initial pressure is 1bar, calculate i) Net heat flow ii) Net change in entropy Represent the process on T-S plot	10	L3	5	2